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TransCatheter aortic valve implantation and fractional flow reserve-guided percutaneous coronary intervention versus conventional surgical aortic valve replacement and coronary bypass grafting for treatment of patients with aortic valve stenosis and complex or multivessel coronary disease (TCW)

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TransCatheter aortic valve implantation and fractional flow reserve-guided percutaneous coronary intervention versus conventional surgical aortic valve replacement and coronary bypass grafting for treatment of patients with aortic valve stenosis and complex or multivessel coronary disease (TCW): an international, multicentre, prospective, open-label, non-inferiority, randomised controlled trial

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Summary

Background Patients with severe aortic stenosis present frequently (~50%) with concomitant obstructive coronary artery disease. Current guidelines recommend combined surgical aortic valve replacement (SAVR) and coronary artery bypass grafting (CABG) as the preferred treatment. Transcatheter aortic valve implantation (TAVI) and fractional flow reserve (FFR)-guided percutaneous coronary intervention (PCI) represent a valid treatment alternative. We aimed to test the non-inferiority of FFR-guided PCI plus TAVI versus SAVR plus CABG in patients with severe aortic stenosis and complex coronary artery disease.

Methods This international, multicentre, prospective, open-label, non-inferiority, randomised controlled trial was conducted at 18 tertiary medical centres across Europe. Patients (aged ≥70 years) with severe aortic stenosis and complex coronary artery disease, deemed feasible for percutaneous or surgical treatment according to the on-site Heart Team, were randomly assigned (1:1) to FFR-guided PCI plus TAVI or SAVR plus CABG according to a computer-generated sequence with random permuted blocks sizes stratified by site. The primary endpoint was a composite of all-cause mortality, myocardial infarction, disabling stroke, clinically driven target-vessel revascularisation, valve reintervention, and life-threatening or disabling bleeding at 1 year post-treatment. The trial was powered for non-inferiority (with a margin of 15%) and if met, for superiority. The primary and safety analyses were done per an intention-to-treat principle. This trial is registered with ClinicalTrials.gov (NCT03424941) and is closed.

Findings Between May 31, 2018, and June 30, 2023, 172 patients were enrolled, of whom 91 were assigned to the FFR-guided PCI plus TAVI group and 81 to the SAVR plus CABG group. The mean age of patients was 76·5 years (SD 3·9). 118 (69%) of 172 patients were male and 54 (31%) patients were female. FFR-guided PCI plus TAVI resulted in favourable outcomes for the primary endpoint (four [4%] of 91 patients) versus SAVR plus CABG (17 [23%] of 77 patients; risk difference –18·5 [90% CI –27·8 to –9·7]), which was below the 15% prespecified non-inferiority margin ($p_{non\text{-}inferiority}$<0·001). FFR-guided PCI plus TAVI was superior to SAVR plus CABG (hazard ratio 0·17 [95% CI 0·06–0·51]; $p_{superiority}$<0·001), which was driven mainly by all-cause mortality (none [0%] of 91 patients *vs* seven (10%) of 77 patients; p=0·0025) and life-threatening bleeding (two [2%] *vs* nine [12%]; p=0·010).

Interpretation The TCW trial is the first trial to compare percutaneous treatment versus surgical treatment in patients with severe aortic stenosis and complex coronary artery disease, showing favourable primary endpoint and mortality outcomes with percutaneous treatment.

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*The TCW study group are listed in the appendix

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Research in context

Evidence before this study

Patients with severe aortic stenosis present frequently with concomitant obstructive coronary artery disease. Percutaneous coronary intervention (PCI) of advanced multivessel coronary artery disease has drastically improved in the past two decades. PCI is considered an equivalent treatment to coronary artery bypass grafting (CABG) for patients with low SYNTAX score while presenting as a valid alternative in patients with high SYNTAX scores and high surgical risk for CABG. A fractional flow reserve (FFR)-guided approach remains the gold standard revascularisation strategy and has been shown to be effective in patients with aortic stenosis. Transcatheter aortic valve implantation (TAVI) has evolved as an alternative to surgical aortic valve replacement (SAVR) for the treatment of aortic stenosis in patients with intermediate or low surgical risk. Although patients who receive combined SAVR and CABG have a higher risk for detrimental short-term outcomes than if each of these treatments were received alone, this combined treatment strategy has been endorsed by the recommendations of the European Society of Cardiology and European Association for Cardiothoracic Surgery guidelines for the management of valvular heart disease. We searched PubMed, MEDLINE, and Cochrane Central Register of Controlled Trials with the search terms TAVI, PCI, CABG, SAVR and randomised trial for publications published in all languages between Jan 1, 2010, and Jan 1, 2024. To our knowledge, no other trial has compared the direct outcomes of TAVI plus PCI versus SAVR plus CABG in this patient population.

Added value of this study

The Transcatheter Valve and Vessels (TCW) trial is the first trial to compare percutaneous and surgical treatment of aortic stenosis and concomitant complex coronary artery disease. The percutaneous approach resulted in favourable outcomes in terms of a composite of all-cause mortality, myocardial infarction, disabling stroke, clinically driven target vessel revascularisation, valve reintervention, and life-threatening bleeding at 1-year follow-up. Importantly, mortality was lower in the FFR-guided PCI plus TAVI group, a finding which led to early termination of the trial.

Implications of all the available evidence

Although corroboration of these findings from larger randomised trials is required, the outcomes of the TCW trial might serve as a valuable reference to delineate decision making in this high-risk patient population and stimulate further research in this area.

Introduction

Patients with severe aortic stenosis present frequently with concomitant obstructive coronary artery disease with a prevalence of 40–70%.[1,2] Following the European Society of Cardiology and European Association for Cardiothoracic Surgery guidelines for the management of valvular heart disease,[3] combined surgical aortic valve replacement (SAVR) and coronary artery bypass grafting (CABG) represent the first-line treatment in these patients. Combined SAVR plus CABG is associated with significantly higher periprocedural and short-term follow-up mortality and morbidity than SAVR or CABG alone.[4–7]

Transcatheter aortic valve implantation (TAVI) has been shown to be non-inferior or superior to SAVR for the treatment of aortic stenosis in patients with intermediate surgical risk or low surgical risk for short-term to intermediate-term outcome.[8–11] Similarly, the treatment of complex or multivessel coronary artery disease has evolved substantially in the past decade. In patients with a low SYNTAX score,[12] percutaneous coronary intervention (PCI) is considered equivalent to CABG while being a valid treatment alternative for patients with a high SYNTAX score but with a high risk for CABG.[13,14] Furthermore, based on the FAME trial results, current PCI strategies focus on the treatment of lesions causing ischaemia.[15] Fractional flow reserve (FFR) is the gold standard technique to detect ischaemia and guide revascularisation in the catheterisation laboratory.[16] The use of FFR in the setting of aortic stenosis might be challenging. However, a recent study showed that FFR is feasible and reliable in most patients with aortic stenosis.[17]

Considering the increased risk of adverse events with combined surgical SAVR plus CABG, a percutaneous approach for combined aortic stenosis and complex coronary artery disease becomes appealing. Therefore, we performed a prospective, non-inferiority, randomised clinical trial comparing FFR-guided PCI plus TAVI versus conventional SAVR plus CABG in patients with severe aortic stenosis and concomitant complex coronary artery disease.

Methods

Study design

The TCW trial is an international, multicentre, prospective, open-label, non-inferiority, randomised controlled trial conducted in 18 tertiary medical centres across the Netherlands (n=6), Spain (n=2), France (n=2), Poland (n=2), Austria (n=1), Czech Republic (n=1), Germany (n=1), Greece (n=1), Portugal (n=1), and Slovakia (n=1). Ethics committee approval was first obtained in the Netherlands (METC Isala Zwolle, 170807) and subsequently in all participating countries and centres. The design of the trial has been published previously.[18] The study was conducted according to the principles of the Declaration of Helsinki and in accordance with the Medical Research Involving Human Subjects Act and was amended by the Declaration of Helsinki 59th WMA General Assembly, (Seoul, October 2008). This trial is registered with ClinicalTrials.gov (NCT03424941) and is closed.

Participants

Patients were aged 70 years or older presenting with symptomatic severe aortic stenosis and concomitant multivessel or complex coronary artery disease determined feasible for percutaneous intervention (ie, FFR-guided PCI plus TAVI) and conventional surgery (ie, SAVR plus CABG) by the on-site Heart Team (ie, at least one cardiac surgeon and one interventional cardiologist with TAVI experience). Aortic stenosis was defined as severe if one of the following criteria were fulfilled: aortic valve area 1 cm^2 or less, mean transvalvular gradient 40 mm Hg or more, aortic jet velocity less than 4 m/s, or velocity index 0·25 or less. Coronary disease was considered complex in the presence of either two or more de novo coronary lesions (≥50% diameter stenosis on visual estimation) in any native epicardial coronary arteries or large side branches (lumen calibre >2 mm) or a single left anterior descending lesion (>20 mm length) involving a bifurcation with a diagonal branch. All inclusion and exclusion criteria are provided in the appendix (pp 3–4). Sex and gender data were collected from medical reports of referring centres. All participants gave written informed consent.

Randomisation and masking

Patients were randomly assigned (1:1) to receive FFR-guided PCI plus TAVI or SAVR plus CABG using a dedicated website as part of the electronic case report form according to a computer-generated sequence with random permuted blocks sizes stratified by site. Patients, investigators, and members of the clinical event adjudication committee were not masked to group assignment. The schematic trial design is presented in the appendix (p 8).

Procedures

Patients in the FFR-guided PCI plus TAVI group had an FFR assessment with intravenous or intracoronary adenosine at operator's discretion following the FFR protocol of each centre. However, intravenous adenosine was strongly recommended. For lesions with an FFR of 0·80 or less, PCI with a modern drug-eluting stent was undertaken. After PCI, patients underwent TAVI with CoreValve Evolut R (Medtronic, Santa Rosa, CA, USA) or CoreValve Evolut PRO (Medtronic, Santa Rosa, CA, USA). Standard recommendations for commissural alignment as provided by the manufacturer were followed. TAVI could occur simultaneously or during a second procedure that was planned no more than 40 days from index PCI. Follow-up visits were scheduled at 30 days and 1 year after TAVI. Another revascularisation procedure was permitted by protocol in patients who remained symptomatic at 30 days follow-up (ie, FFR values <0·80). At discharge, all patients received optimal medical treatment following the European Society of Cardiology guidelines.

In the SAVR plus CABG group, patients received CABG plus SAVR with a bioprosthetic aortic valve (Hancock or Hancock Ultra, Medtronic, USA; Perimount Magna Ease, Edward Lifesciences, USA; Trifecta, Abbott, USA; Perceval, LivaNova, UK). FFR guidance was not mandatory and was left to the surgeon's discretion. Use of arterial grafts was strongly recommended. Patients received optimal medical treatment at discharge, and follow-up was scheduled at 30 days and 1 year during an in-person visit.

Outcomes

The primary endpoint was a patient-oriented composite of all-cause mortality, myocardial infarction, disabling stroke, unscheduled clinically driven target vessel revascularisation, valve reintervention, and life-threatening or disabling bleeding at 1 year. The two major secondary endpoints were major adverse cardiovascular events comprising cardiovascular mortality, all stroke, myocardial infarction, unscheduled coronary events, or valve reintervention at 1 year, as well as a composite of all-cause mortality and stroke at 1 year. A detailed list of all endpoints and their definitions is described elsewhere.[18] The endpoint definitions are based on the Valve Academic Research Consortium-2 initiative criteria.[19]

Adverse events were adjudicated by an independent clinical event committee comprising of two interventional cardiologists and two cardiac surgeons. The echocardiogram data reported from the participating centres were re-evaluated from an independent core laboratory

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See **Online** for appendix

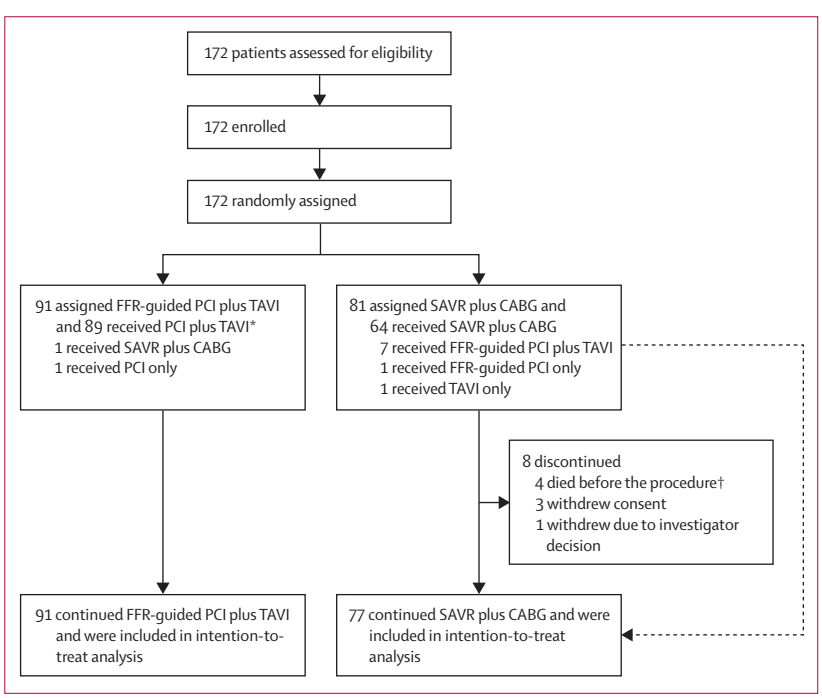

Figure 1: **Trial profile**
CABG=coronary artery bypass grafting. FFR=fractional flow reserve. PCI=percutaneous coronary intervention. SAVR=surgical aortic valve replacement. TAVI=transcatheter aortic valve implantation. *73 received FFR-guided PCI plus TAVI. †As mandated per protocol, these four patients were not included in the intention-to-treat analysis.

	FFR-guided PCI plus TAVI (n=91)	SAVR plus CABG (n=81)
Age, years	76·3 (3·7)	76·6 (4·1)
70–80	77/91 (85%)	64/81 (79%)
>80	14/91 (15%)	17/81 (21%)
Sex		
Men	59/91 (65%)	59/81 (73%)
Women	32/91 (35%)	22/81 (27%)
Ethnicity		
White	80/91 (88%)	73/81 (90%)
Black	1/91 (1%)	0
Hispanic, Latino, Latina, or Latinx	6/91 (7%)	4/81 (5%)
Other	4/91 (4%)	4/81 (5%)
Cardiac risk factors		
Family history	20/89 (23%)	24/77 (31%)
Hyperlipidaemia	57/91 (63%)	48/78 (62%)
Smoking	7/91 (8%)	12/77 (16%)
Hypertension	69/91 (76%)	65/78 (83%)
Previous myocardial infarction	10/91 (11%)	8/78 (10%)
Previous percutaneous coronary intervention	13/91 (14%)	4/78 (5%)
Known history of atrial fibrillation or atrial flutter	17/91 (19%)	15/78 (19%)
Diabetes	27/91 (30%)	27/78 (35%)
Non-insulin dependent	23/91 (25%)	19/78 (24%)
Insulin dependent	4/91 (4%)	8/78 (10%)
Chronic renal failure (GFR <60 mL/min per m^2)	15/91 (16%)	15/78 (19%)
Chronic lung disease	10/91 (11%)	8/78 (10%)
Previous cerebrovascular incident	5/91 (5·5%)	5/78 (6%)
Known history of arrhythmia	18/91 (20%)	19/78 (24%)
Peripheral vascular disease	9/91 (10%)	10/78 (13%)
Angina Pectoris Canadian Cardiovascular Society class		
I	52/87 (60%)	41/74 (55%)
II	21/87 (24%)	21/74 (28%)
III	13/87 (15%)	9/74 (12%)
IV	1/87 (1%)	3/74 (4%)
New York Heart Association class		
I	18/91 (20%)	20/76 (26%)
II	42/91 (46%)	32/76 (42%)
III	29/91 (32%)	23/76 (30%)
IV	2/91 (2%)	1/76 (1%)

(Table 1 continues in next column)

	FFR-guided PCI plus TAVI (n=91)	SAVR plus CABG (n=81)
(Continued from previous column)		
EuroSCORE II	2·3% (1·7–3·3)	2·8% (1·9–3·9)
STS-PROM score	2·3 (1·7–3·6)	2·6 (1·7–3·9)
Low risk (<4%)	70/90 (78%)	59/77 (77%)
Intermediate risk (4–8%)	17/90 (19%)	14/77 (18%)
High risk (>8%)	3/90 (3%)	4/77 (5%)
SYNTAX score	13·0 (10·0–17·0)	12·0 (8·0–17·0)
Low (≤16)	57/82 (70%)	56/75 (75%)
Intermediate (16–22)	14/82 (17%)	12/75 (16%)
High (≥22)	11/82 (13%)	7/75 (9%)
Number of diseased vessels (>50%)		
One vessel	21/91 (23%)	20/78 (26%)
Two vessels	46/91 (51%)	41/78 (53%)
Three or more vessels	23/91 (25%)	18/78 (23%)
Lesion distribution		
Left main	2/89 (2%)	3/76 (4%)
Left anterior descending artery	77/89 (87%)	59/76 (78%)
Right coronary artery	46/89 (52%)	40/76 (53%)
Circumflex artery	44/89 (49%)	36/76 (47%)
Aortic valve		
Mean gradient, mm Hg	43 (12·1)	43·6 (12·1)
Maximum gradient, mm Hg	67·3 (60·0–80·5)	71·7 (57·8–86·3)
Maximum velocity, m/s	4·2 (3·9–4·6)	4·2 (3·8–4·6)
Aortic valve area,* cm^2	0·6 (0·5–0·7)	0·5 (0·4–0·7)
Velocity index	0·2 (0·2–0·3)	0·2 (0·2–0·3)
LVEF, %	54·6 (9·4)	54·4 (8·8)
Edmonton Frail Scale, points	3·3 (2·2)	3·5 (2·0)

Data are n (%), mean (SD), or median (IQR). CABG=coronary artery bypass grafting. FFR=fractional flow reserve. GFR=glomerular filtration rate. LVEF=left ventricular ejection fraction. PCI=percutaneous coronary intervention. SAVR=surgical aortic valve replacement. STS-PROM=Society of Thoracic Surgeons-Predicted Risk of Mortality. TAVI=transcatheter aortic valve implantation. *Aortic valve area was determined by velocity time integral.

Table 1: **Baseline characteristics**

(Diagram BV, Zwolle, Netherlands) in a sample representing a third of the total patients (107 patients).

Statistical analysis

Assuming a primary endpoint rate of 30% in the SAVR plus CABG group, with a one-sided α of 0·05 and a non-inferiority margin of 15%, a total of 320 patients (160 per group) were needed to obtain 90% power. Considering a 2% loss to follow-up, 328 patients in total were needed for enrolment. On June 30, 2023, enrolment was stopped as a safety data analysis (done in 33% of patients and repeated at 50% of enrolment *vs* 66% in the protocol) showed an important difference between the two groups (p<0·0001) for the primary endpoint and all-cause mortality. Therefore, 172 patients were recruited in total and were followed up at 1 year.

The primary endpoint was analysed per intention to treat, which included events from the day of the (first) procedure. The safety analysis was done per intention to treat, which included all events starting from enrolment. A per-protocol analysis and as-treated analysis were performed as supportive evidence. The per-protocol analysis included only patients from each group who did not deviate from the protocol, whereas the as-treated analysis included patients who received percutaneous or surgical treatment independently from their original assignment group.

Following the protocol, non-inferiority was met if the upper bound of the 90% CI of the difference between percutaneous treatment and surgical treatment was below 15%. If non-inferiority was met, a superiority analysis with 95% CIs would be performed. When the upper confidence limit of the hazard ratio (HR) was less than 1, superiority of FFR-guided PCI plus TAVI over SAVR plus CABG would be claimed. As both tests were disjunct, no correction for α-spending was required. Event rates were based on Kaplan–Meier estimates in time-to-first-event analyses. Non-inferiority was calculated using the Com-Nougue approach to estimate the Z statistic for Kaplan–Meier failure rates, with standard errors estimated by using the means of Greenwood's formula. In time-to-first-event analyses, HRs with 95% CIs were calculated and event rates were compared using the log-rank test. The same methods were used for the analysis of secondary endpoints. Categorical data for baseline and procedural characteristics were summarised by number of patients and total percentages, and continuous data were summarised by means, standard deviations, median, and range according to data distribution. Categorical variables were compared using the χ^2 test or Fisher's exact test. Continuous variables were compared using Student's *t* test or the Wilcoxon rank-sum test for not normally distributed data. For superiority, a two-sided p value of 0·05 or less was considered statistically significant. All statistical analyses were performed using SAS, version 9.4. An independent data safety monitoring committee was used, which comprised of three members, one of whom was an experienced biostatistician who held regular meetings during the trial (appendix p 2).

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Between May 31, 2018, and June 30, 2023, 172 patients were enrolled, of whom 91 were randomly assigned to the FFR-guided PCI plus TAVI group and 81 to the SAVR plus CABG group (figure 1). During the COVID-19 pandemic, based on a joint decision by the steering committee and data safety monitoring board, enrolment was paused in all participating centres on May 13, 2020, until reinitiating enrolment was considered feasible and safe. Of 81 patients in the SAVR plus CABG group, four died before the procedure and were not included in the intention-to-treat analysis. No patients were lost to follow-up.

Baseline characteristics were well-balanced between the groups (table 1). The mean age of patients was 76·5 years (SD 3·9). 118 (69%) of 172 patients were male and 54 (31%) patients were female. Cardiac risk factors were well-balanced between the groups, including the New York Heart Association class and angina class (Canadian Cardiovascular Society classification). The median EuroSCORE II was similar between the

	FFR-guided PCI plus TAVI (n=89)	SAVR plus CABG (n=64)
Lesions per patient	2·2 (1·0)	2·3 (0·8)
FFR-guided revascularisation	73 (82%)	19 (30%)
Lesions treated per patient	1·5 (0·6)	NA
Complete revascularisation	74 (81%)	41 (64%)
Total stent length, mm	53·7 (29·4)	NA
Procedural success	88 (99%)	NA
Femoral access	84 (94%)	NA
Subclavian access	5 (6%)	NA
Conscious sedation	63 (71%)	NA
Successful implantation	89 (100%)	NA
Valve migration or embolisation	0	NA
Device size, mm	29·7 (3·0)	23·5 (2·15)
Device success	89 (100%)	NA
Vessels grafted	NA	1·6 (0·8)
Arterial grafts only	NA	27 (42%)
Arterial and venous grafts	NA	25 (39%)
Venous grafts only	NA	12 (19%)
Biological aortic valve	NA	64 (100%)
Hancock or Hancock Ultra (Medtronic, USA)	NA	10 (16%)
Perimount Magna Ease (Eduard Lifesciences, USA)	NA	41 (64%)
Trifecta (Abbott, USA)	NA	10 (16%)
Parcival (Livanova, UK)	NA	3 (5%)

Data are n (%) or mean (SD). CABG=coronary artery bypass grafting. FFR=fractional flow reserve. NA=not applicable. PCI=percutaneous coronary intervention. SAVR=surgical aortic valve replacement. TAVI=transcatheter aortic valve implantation.

Table 2: **Patient-level procedural data in the per-protocol population**

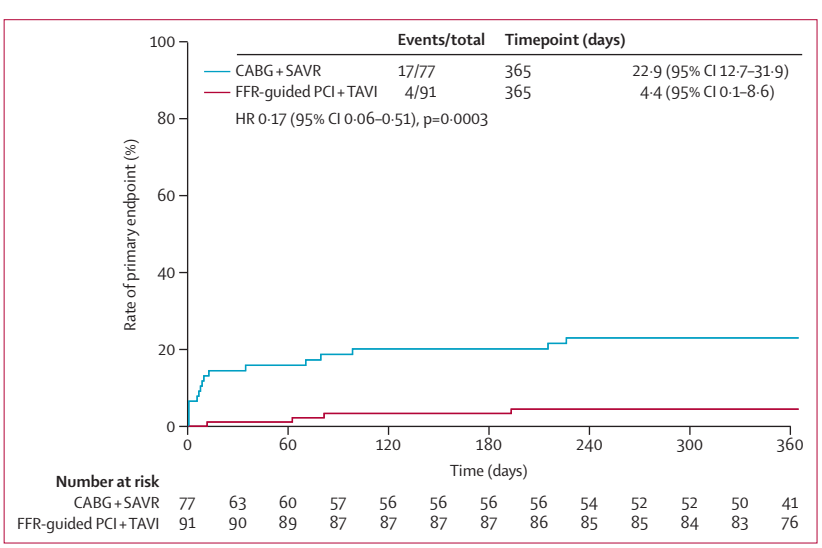

Figure 2: **Kaplan–Meier plot of the primary endpoint in the intention-to-treat population**
CABG=coronary artery bypass grafting. FFR=fractional flow reserve. HR=hazard ratio. PCI=percutaneous coronary intervention. SAVR=surgical aortic valve replacement. TAVI=transcatheter aortic valve implantation.

FFR-guided PCI plus TAVI group (2·3% [IQR 1·7–3·3]) and the SAVR plus CABG group (2·8% [1·9–3·9]). Overall, most patients presented with low-risk and intermediate-risk mortality according to the Society of Thoracic Surgeons-Predicted Risk of Mortality (STS-PROM) score, with only seven (4%) of 167 patients having a high score. The median STS-PROM and EuroSCORE II scores were 2·4 for the whole trial population. The overall median SYNTAX score was 12·0 (IQR 9·0–17·0) and only 44 (28%) of 157 patients had an intermediate or high SYNTAX score with 128 (76%) of 169 patients presenting with multivessel coronary artery disease (two-or-more vessel disease). The median time from enrolment to procedure was higher in the SAVR plus CABG group (53 days) than in the FFR-guided PCI plus TAVI group (19 days).

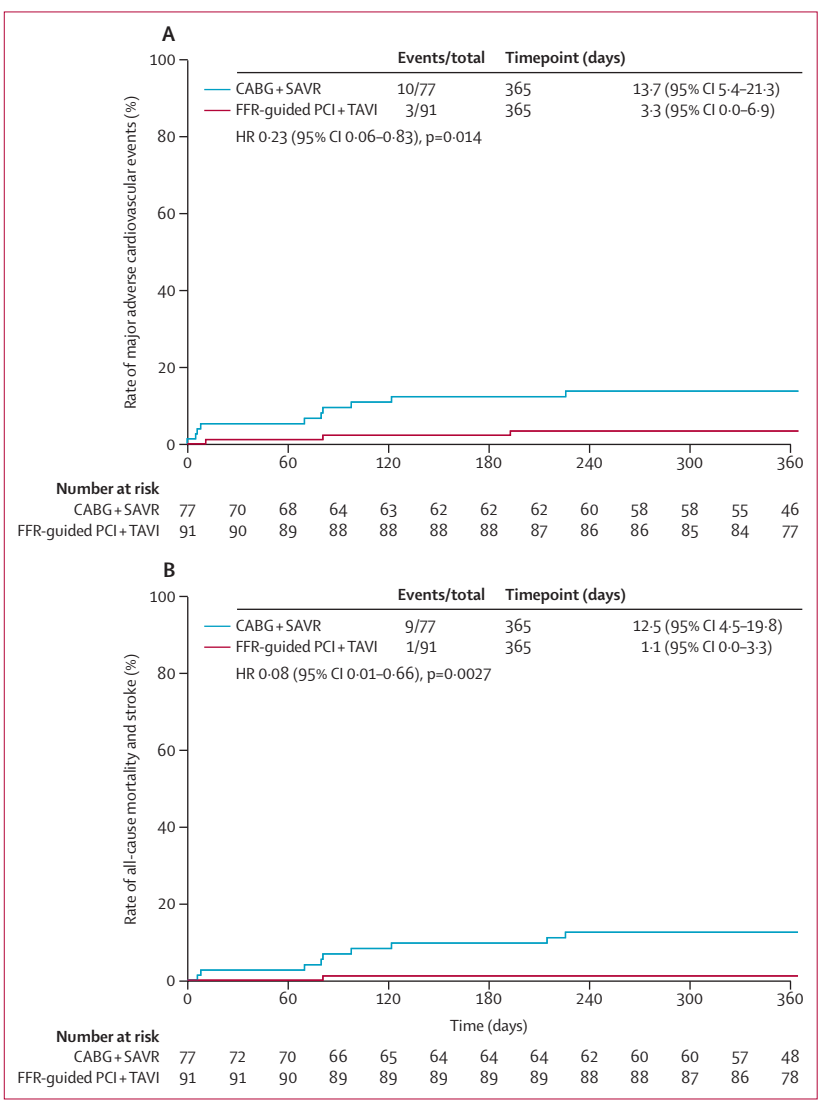

***Figure 3*: Kaplan–Meier plot of major adverse cardiac events (A) and all-cause mortality and stroke (B) at 1 year in the intention-to-treat population**

Major adverse cardiovascular events were a composite of cardiovascular mortality, all stroke, myocardial infarction, unscheduled coronary, or valve reintervention at 1 year. The other main secondary endpoint was composite of all-cause mortality and stroke at 1 year. CABG=coronary artery bypass grafting. FFR=fractional flow reserve. HR=hazard ratio. PCI=percutaneous coronary intervention. SAVR=surgical aortic valve replacement. TAVI=transcatheter aortic valve implantation.

In the FFR-guided PCI plus TAVI group, 73 (82%) of 89 patients that underwent PCI had FFR-guided PCI. The patients that did not undergo FFR assessment often had a single but severe lesion in the left anterior descendent artery or had angiographically diffuse tortuous lesions not suitable for FFR. In some centres, the FFR procedure was temporarily not possible due to material shortages during the COVID-19 pandemic. The mean number of grafts used (ie, lesions treated) per patient in the FFR-guided PCI group was 1·5 (SD 0·6) versus 1·6 (SD 0·8) in the SAVR plus CABG group (table 2). Success of PCI was reported in 88 (99%) of 89 patients in the FFR-guided PCI plus TAVI group and device success was reported in all patients treated with TAVI. In the FFR-guided PCI plus TAVI group, the protocol permitted further revascularisation if angina persisted at 30-day follow-up; however, this was not the case in any of the patients. In the SAVR plus CABG group, an average of 1·6 vessels (SD 0·8) were grafted. Revascularisation with only arterial grafts, which was strongly recommended, was achieved in 27 (42%) of 64 patients, whereas the remaining 37 (58%) patients underwent CABG with mixed arterial and venous or venous only grafts. A biological prosthesis was used in all patients that underwent SAVR.

At 1 year, FFR-guided PCI plus TAVI resulted in favourable outcomes for the primary endpoint (four [4%] of 91 patients) compared with SAVR plus CABG (17 [23%] of 77 patients; risk difference –18·5 [90% CI –27·8 to –9·7]; figure 2). The upper bound of the 90% two-sided CI of the difference between the two groups was –9·7% and, therefore, was below 15%, confirming non-inferiority between the groups ($p_{non\text{-}inferiority}$<0·001). Since non-inferiority was met, superiority was tested. FFR-guided PCI plus TAVI treatment was found to be superior to SAVR plus CABG (HR 0·17 [95% CI 0·06–0·51]; $p_{superiority}$<0·001). The main drivers were all-cause mortality, which was lower in the FFR-guided plus PCI group (none [0%] of 91 patients) than in the SAVR plus CABG group (seven [10%] of 77 patients; p=0·0025), and life-threatening or disabling bleeding (two [2%] of 91 *vs* nine [12%] of 77; HR 0·17 [95% CI 0·04–0·80]; p=0·010). Six of seven patients died from cardiovascular causes, most of which were directly related to the procedure (n=5). Only one of these deaths occurred within 30 days, whereas the majority of cardiovascular-related deaths occurred within the first 1–4 months after the procedure. The safety analysis, per-protocol analysis, and as-treated analysis outcomes are presented in the appendix (pp 5–7, 9–11).

The rate of major adverse cardiac events at 1 year was lower in the FFR-guided PCI plus TAVI group (three [3%] of 91 patients) than in the SAVR plus CABG group (ten [14%] of 77 patients; HR 0·23 [95% CI 0·06–0·83]; p=0·014; figure 3A). Similarly, a composite of all-cause

mortality and any stroke rate was lower in the FFR-guided PCI plus TAVI group (one [1%] of 91 patients) than in the SAVR plus CABG group (nine [12%] of 77 patients; HR 0·08 [95% CI 0·01–0·66]; p=0·0027; table 3; figure 3B). The hospitalisation duration was shorter in the FFR-guided PCI plus TAVI group (7·0 days) than in the SAVR plus CABG group (9·9 days; p=0·0058). As expected, the rate of pacemaker implantation was higher in the FFR-guided PCI plus TAVI group (nine [10%] of 91 patients) than in the SAVR plus CABG group (two [3%] of 77 patients; HR 3·74 [95% CI 0·81–17·30]; p=0·070). By contrast, the SAVR plus CABG group had a higher rate of atrial fibrillation (four [4%] *vs* 11 [14%]; HR 0·28 [95% CI 0·09–0·88], p=0·020) and rethoracotomy (none [0%] *vs* four [5·2%]; p=0·028) than the FFR-guided PCI plus TAVI group. Major vascular complications did not differ significantly between the groups.

Discussion

The TCW trial is the first international, prospective, non-inferiority, randomised controlled trial comparing percutaneous treatment (FFR-guided PCI plus TAVI) with surgical treatment (SAVR plus CABG) in patients with symptomatic aortic stenosis and concomitant complex coronary artery disease. Percutaneous treatment showed favourable outcomes compared with surgical treatment. The outcomes of the intention-to-treat analysis were further corroborated by the findings of the per-protocol analysis, safety analysis, and as-treated analysis. Importantly, higher rates of major adverse cardiac events, all-cause mortality, and stroke, as well as cardiac mortality occurred in the SAVR plus CABG group than in the FFR-guided PCI plus TAVI group.

This trial enrolled patients aged 70 years and older with low and moderate mortality risk based on STS-PROM and EuroSCORE II evaluation since patients with a high mortality risk are not accepted for surgery. The median STS-PROM and ESII scores were 2·4 for the entire trial population, which is in line with scores from the German Aortic Valve Registry report,[20] which is one of the largest European reports on patients with aortic stenosis or coronary artery disease. A subanalysis of the SURTAVI trial,[21] in which the same self-expanding TAVI device was used, showed a slightly higher STS-PROM score in the same patient population. However, since STS-PROM scores fluctuate over time[22] (read reduction), our scores would also likely have been marginally higher if they had been calculated at the same time as the SURTAVI trial. The German Aortic Valve Register and the SURTAVI trial reported higher 30-day mortality rates in patients undergoing combined SAVR plus CABG than patients in our trial. In SURTAVI, the 1-year mortality and stroke rate was lower than the rate reported in our trial. One possible reason for this finding is that patients in our trial had more extensive coronary disease. Indeed, the reported SYNTAX score was higher in our trial than in

	FFR-guided PCI plus TAVI (n=91)	SAVR plus CABG (n=77)	HR (95% CI)	p value
Clinical outcomes at 30 days				
Primary endpoint	1 (1%)	11 (14%)	0·07 (0·01–0·55)	0·0008
Secondary endpoint				
Major adverse cardiac events	1 (1%)	4 (5%)	0·20 (0·02–1·82)	0·11
All-cause mortality and stroke	0	2 (2%)	NA	0·12
Death—all cause	0	1 (1%)	NA	0·28
Death—cardiovascular	0	1 (1%)	NA	0·28
Stroke or TIA	0	2 (3%)	NA	0·12
Disabling stroke	0	1 (2%)	NA	0·27
Non-disabling stroke	0	0	NA	NA
TIA	0	1 (1%)	NA	0·27
Myocardial infarction (any)	1 (1%)	1 (1%)	0·82 (0·05–13·18)	0·89
Periprocedural myocardial infarction	1 (1%)	1 (1%)	0·82 (0·05–13·18)	0·89
Spontaneous myocardial infarction	0	0	NA	NA
Any revascularisation	0	1 (1%)	NA	0·28
CD-TVR	0	1 (1%)	NA	0·28
Valve reintervention	0	1 (1%)	NA	0·28
Life threatening or disabling bleeding (VARC-2)	1 (1%)	7 (9%)	0·11 (0·01–0·93)	0·014
Major bleeding (VARC-2)	2 (2%)	6 (8%)	0·28 (0·06–1·39)	0·091
Minor bleeding (VARC 2)	8 (9%)	3 (4%)	2·24 (0·59–8·44)	0·22
Permanent pacemaker implantation	8 (9%)	1 (1%)	6·74 (0·84–53·88)	0·037
Major vascular complication	4 (4%)	1 (1%)	3·36 (0·38–30·09)	0·25
Rethoracotomy	0	4 (5%)	NA	0·028
Atrial fibrillation	2 (2%)	10 (13%)	0·16 (0·03–0·72)	0·006
Clinical outcomes at 1 year				
Primary endpoint	4 (4%)	17 (23%)	0·17 (0·06–0·51)	$p_{non\text{-}inferiority}$ <0·001; $p_{superiority}$ <0·001
Secondary endpoint				
Major adverse cardiac events	3 (3%)	10 (14%)	0·23 (0·06–0·83)	0·014
All-cause mortality and stroke	1 (1%)	9 (12%)	0·08 (0·01–0·66)	0·0027
Death—all cause	0	7 (10%)	NA	0·0025
Death—cardiovascular	0	6 (8%)	NA	0·0052
All stroke and TIA	1 (1%)	3 (4%)	0·26 (0·03–2·46)	0·20
Disabling stroke	1 (1%)	2 (3%)	0·38 (0·03–4·22)	0·42
Non-disabling stroke	0	0	NA	NA
TIA	0	1 (1%)	NA	0·27
Myocardial infarction (any)	2 (2%)	1 (1%)	1·58 (0·14–17·48)	0·71
Periprocedural myocardial infarction	1 (1%)	1 (1%)	0·82 (0·05–13·18)	0·89
Spontaneous myocardial infarction	1 (1%)	0	NA	0·39
Any revascularisation	0	1 (1%)	NA	0·28
CD-TVR	0	1 (1%)	NA	0·28
Valve reintervention	0	1 (1%)	NA	0·28
Life threatening or disabling bleeding (VARC-2)	2 (2%)	9 (12%)	0·17 (0·04–0·80)	0·010
Major bleeding (VARC-2)	5 (6%)	7 (9%)	0·57 (0·18–1·79)	0·32

(Table 3 continues on next page)

	FFR-guided PCI plus TAVI (n=91)	SAVR plus CABG (n=77)	HR (95% CI)	p value
(Continued from previous page)				
Minor bleeding (VARC-2)	12 (13%)	4 (5%)	2·52 (0·81–7·81)	0·10
Permanent pacemaker implantation	9 (10%)	2 (3%)	3·74 (0·81–17·30)	0·070
Major vascular complication	4 (4%)	1 (1%)	3·36 (0·38–30·09)	0·25
Rethoracotomy	0	4 (5%)	NA	0·028
Atrial fibrillation	4 (4%)	11 (14%)	0·28 (0·09–0·88)	0·020

Data are n (%). Major adverse cardiovascular events were defined as cardiovascular mortality, all stroke, myocardial infarction, unscheduled coronary, or valve reintervention. CABG=coronary artery bypass grafting. CD-TVR=clinical driven target vessel revascularisation. FFR=fractional flow reserve. HR=hazard ratio. NA=not applicable. PCI=percutaneous coronary intervention. SAVR=surgical aortic valve replacement. TAVI=transcatheter aortic valve implantation. TIA=transient ischaemic attack. VARC-2=Valve Academic Research Consortium-2.

Table 3: **Clinical outcomes at 30 days and 1 year**

SURTAVI (13·4 *vs* 8·5). As shown from the German Aortic Valve Registry,[20] the duration of the procedure is directly related to worse outcomes regarding mortality and life-threatening bleeding post-surgery. SAVR plus CABG surgery was 1 h longer on average than isolated SAVR and was associated with a two-fold increase in mortality and bleeding. Therefore, patients with more advanced coronary disease require longer operations that can result in worse outcomes, as observed in our trial. By contrast, intra-procedural complications are associated with longer surgical interventions and increased adverse events. Another important finding was that the 30-day mortality rate was in line with the STS-PROM score. However, five of six cardiovascular deaths were directly related to post-surgical treatment complications that occurred between 30 days and 90 days. This finding probably reflects the developments in intensive care medicine and improved post-surgical patient care, which manage post-procedure complications beyond 30 days. Therefore, scores capturing longer follow-ups (ie, up to 90 days, post-cardiac surgery, or intervention) might be more appropriate than an STS-PROM score which is limited to 30 days.

The event rate in the FFR-guided PCI plus TAVI group was low, which could reflect improved outcomes over time in patients undergoing PCI plus TAVI. As TAVI has become a routine procedure in catheterisation laboratories in Europe, and as the risk of patients treated with TAVI has diminished, the clinical outcomes of TAVI have drastically improved. Indeed, the rate of the primary endpoint in our trial is similar to that observed in the Evolute Low Risk trial.[11]

The incidence of life-threatening and disabling bleeding was higher in the SAVR plus CABG group. This finding, as well as the observed rate of bleeding, is in line with other studies.[10,20] In the German Aortic Valve Register, 28% of patients undergoing combined SAVR and CABG required blood transfusions of 4 units or more versus 16% of patients undergoing isolated SAVR, indicating that blood transfusions during surgery are common with combined valve and coronary surgery. Although life-threatening or disabling bleeding was a composite of the primary endpoint, the other main secondary endpoints, particularly major adverse cardiac events, excluding the impact of bleeding, also resulted in lower event rates in patients who received PCI, with results mainly driven by cardiovascular mortality. Notwithstanding, general bleeding and life-threatening bleeding have a major impact on mortality,[23] as was also the case in our trial.

The TCW trial applied an FFR-guided revascularisation strategy, which might likely result in a more complete revascularisation strategy than in the SURTAVI trial. The FAME 3 trial and our trial showed that FFR-guided revascularisation was associated with a trend towards better outcome than CABG in patients with a low SYNTAX score (<22).[13]

Although the need for coronary revascularisation after successful TAVI is being investigated in a large randomised trial (NCT04634240), complete coronary revascularisation might be associated with better clinical outcomes than partial revascularisation, as shown in the FIRE trial.[24] Furthermore, a medical treatment strategy for patients with ischaemia and multivessel disease was associated with an increased risk of spontaneous myocardial infarction (10%), as shown in the ISCHEMIA trial.[25] Finally, adenosine use during FFR assessment in our trial was shown to be safe, which further supports previous findings.[26]

The TCW trial has some limitations. Firstly, the number of patients enrolled was reduced due to early termination, which can lead to type I and type II errors. Therefore, the biases that can arise from these errors should be considered when interpreting the outcomes. Secondly, only one TAVI device was used (Evolut R or Evolut PRO); therefore, the results cannot be generalised to patients treated with other device types. Thirdly, the use of FFR-guidance in the FFR-guided PCI plus TAVI group was per operator's discretion and, to date, evidence-based data favouring this approach are absent. The validity of FFR in the setting of aortic stenosis is debatable as recent studies have shown a variable effect of aortic stenosis treatment on post-FFR values.[27,28] Multiple factors can affect the FFR value in the setting of aortic stenosis, leading to cases of false negatives or false positives. However, despite these findings, studies agree that haemodynamic assessments, when used cautiously, can provide valid information regarding coronary stenosis severity. We believe that these variations in post-FFR values had little or no effect on the primary outcomes as they were predominantly driven by mortality related complications, surgical complications, and life-threatening bleeding whereas the rate of revascularisation or myocardial infarction was low in both groups. A further limitation is the higher rates of complete revascularisation in the FFR-guided group than in the SAVR plus CABG group. However, this finding should be interpreted only as a methodical-derived

difference as complete revascularisation was calculated from FFR-positive lesions in the FFR-guided PCI plus TAVI group versus all lesions in the SAVR plus CABG group. Nevertheless, the number of revascularised lesions per patient was similar between the groups (table 3). A further limitation was the low proportion of patients with a high SYNTAX score; however, this might reflect the true incidence of patients with advanced coronary disease and concomitant severe aortic stenosis as the presence of aortic stenosis exacerbates coronary symptoms and might lead to an earlier presentation. Additionally, the COVID-19 pandemics caused a transitory halt in enrolment that differed by centre. However, the clinical event committee ruled out a possible effect of COVID-19 infection in the clinical events reported in the intention-to-treat analysis. Nevertheless, the induced dysfunction of health institutions (eg, reduced capacity of intensive care units) in Europe could have increased the time to first procedure, particularly affecting waiting list times for patients enrolled to surgery. This might have affected the safety analysis results, in which four patients assigned to surgery died before the procedure. Another limitation is the high rate of crossover to the FFR-guided PCI plus TAVI group, which could have been partly influenced by a negative FFR assessment in one or more lesions reducing the need for extended surgery, as well as the patient preference for a less invasive procedure irrespective of previous treatment experiences.[29] However, the as-treated analysis supports the findings of the intention-to-treat analysis suggesting that this crossover rate did not affect the main findings. Finally, no mortality was observed in the FFR-guided PCI plus TAVI group, which supports the need for larger trials with longer follow-up.

In conclusion, the TCW trial shows that FFR-guided PCI plus TAVI compared with SAVR plus CABG resulted in favourable outcomes for patient-oriented composite of all-cause mortality, myocardial infarction, disabling stroke, unscheduled clinically driven target vessel revascularisation, valve reintervention, and life-threatening or disabling bleeding, as well as major adverse cardiovascular events and all-cause mortality and stroke. These results were mainly driven by a reduction of all-cause mortality and cardiovascular mortality rates in the patients who received PCI. Although these results bring new insights to the treatment of aortic stenosis and concomitant coronary artery disease, corroboration from larger randomised trials is required.

Contributors

EKe designed the trial, wrote the protocol, and drafted the manuscript. RSH, J-HED, SKS, WW, and TM refined the final protocol. JMTB, DvG, MH, GA, IJA-S, MA, RCT, GB, EVB, LC, LvG, VV, JS, PC, EL-SS, LN-F, SP, KP, GDL, EKo, and KPM interpreted the data and wrote the final manuscript. EKe, TM, and WW verified the data. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

EKe reports institutional research grants from Abbott Vascular Laboratories and Medtronic and proctor or lecture fees from Abbott Vascular. RSH reports lecture fees from Abbott Vascular, Amgen, Novartis, Edwards Lifesciences, and Meril. DvG reports lecture fees from Cardiovasculair Onderwijs Instituut. GA reports lecture fees from Medtronic, Abbott, and Meril Life, patents planned for Tdaflo, and being a data and safety monitoring board member for Vivasure. IJA-S reports proctor, lecture, and consultant fees from Medtronic, Boston Scientific, Meril Life, and Products&Features. MA reports institutional research grants from Edwards, Abbott, Medtronic, and LSI, and proctor, lecture, and consultant fees from Edwards Lifesciences, Abbott, Medtronic, Boston, Zoll, and BBraun. GB reports a consulting grant from Medtronic. LC reports consulting fees from Medtronic, Abbott, JenaValve, Edwards Lifesciences, BostonScientific, PiCardia, MicroPort, VenusMedtech, 4C Medical, Smartcanula, MicroInterventions, and Neovasc, support for attending meetings or travel (or both) from Medtronic, Abbott, JenaValve, Edwards Lifesciences, and BostonScientific, and being a data and safety monitoring board member for Medtronic, Abbott, and JenaValve. LvG reports proctor fees from Edwards Lifesciences and support for attending meetings or travel (or both) from Edwards Lifesciences, Corcym, and Atricure. WW reports lecture fees from Medtronic and is a data safety monitoring board member for Medtronic. VV reports consulting fees from Medtronic. JS reports lecture fees from Abiomed. EL reports institutional research grants from Abbott. LN-F reports consulting fees from Abbot, Edwards Lifesciences, and Boston, and proctor, lecture, and consultant fees from Abbott, Edwards Lifesciences, Medtronic, and Boston. All other authors declare no competing interests.

Data sharing

Data collected for this study will not be made publicly available. However, we will consider collaboration requests with other investigators who wish to perform additional analyses from the study database. On conditional approval from the steering committee the data requested for that analysis will be shared after signing a data agreement. Any such request can be directed to elvin.kedhi@mcgill.ca.

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